Putnam

Hidden independence and uniformity

Written by a model for Adit. Not Adit's words.

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1 Hidden independence and uniformity

Recognise it. The problem has a symmetry it does not mention. Points are chosen at random and the question is about their order or their arrangement. A sum runs over all permutations, all subsets, or all residues, and the summand is built from one coordinate. A quantity is asked for that, once computed, turns out not to depend on a parameter the problem made a fuss about. A sign pattern or a weight pairs terms off against each other. The answer to the smallest case is suspiciously clean: a probability of $1/2$ or $1/n$, an integer where a fraction was expected, a sum that is zero.

The model has three pieces. Uniformity: when a random object is drawn from a set that a bijection maps to itself, the image is drawn from the same distribution, so anything invariant under the bijection can be computed in the most convenient representative. Linearity: an expectation of a sum is a sum of expectations, whether or not the parts are independent, so a complicated count becomes a sum of probabilities of simple events. Independence proper: when a joint distribution factors, a probability of a conjunction is a product, and one hard event becomes several easy ones.

Definition 1.1. A random variable X taking values in a finite set S is uniform if $\Pr(X = s) = 1/|S|$ for every $s \in S$. Events $A_1, \dots, A_n$ are independent if $\Pr(A_{i_1} \cap \cdots \cap A_{i_k}) = \Pr(A_{i_1}) \cdots \Pr(A_{i_k})$ for every choice of distinct indices; pairwise independence is the case $k = 2$ only.

Lemma 1.2 (Transport of uniformity)**.** *Let X be uniform on a finite set S and let $\varphi \colon S \to S$ be a bijection. Then $\varphi(X)$ is uniform on S, and for every function g on S, $\mathbb{E}\, g(X) = \mathbb{E}\, g(\varphi(X))$.*

Proof. $\Pr(\varphi(X) = s) = \Pr(X = \varphi^{-1}(s)) = 1/|S|$. The second claim is the first one applied to the distribution of g. □

Lemma 1.3 (Equal fibres)**.** *Let a finite group G act on a finite set X, let A be a finite abelian group, and let $f \colon X \to A$ and a surjective homomorphism $\chi \colon G \to A$ satisfy $f(g \cdot x) = f(x) + \chi(g)$ for all g, x. Then every fibre $f^{-1}(a)$ has the same size $|X|/|A|$.*

Proof. For $a, b \in A$ pick g with $\chi(g) = b - a$. Then $x \mapsto g \cdot x$ maps $f^{-1}(a)$ into $f^{-1}(b)$, and $x \mapsto g^{-1} \cdot x$ maps it back, so the two fibres are in bijection. □

The lemma is the formal version of "shift everything by one and the sum shifts by a unit". It is why the sum of n fair dice is uniform modulo 6, why a residue class of a symmetric count is as large as any other, and why so many answers are $1/p$.

Theorem 1.4 (Linearity of expectation)**.** *For random variables $X_1, \dots, X_n$ with finite expectations, $\mathbb{E}(X_1 + \cdots + X_n) = \mathbb{E}X_1 + \cdots + \mathbb{E}X_n$. In particular, if N counts how many of the events $A_1, \dots, A_n$ occur, then $\mathbb{E}N = \sum_i \Pr(A_i)$.*

Proof. Expectation is a sum (or integral) weighted by the probability measure, and sums commute with sums. For the second statement write $N = \sum_i \mathbf{1}_{A_i}$, where the indicator $\mathbf{1}_A$ is 1 on A and 0 elsewhere, and use $\mathbb{E}\,\mathbf{1}_A = \Pr(A)$. □

Proposition 1.5 (Exchangeability)**.** *Let $X_1, \dots, X_n$ be independent with the same continuous distribution. Then every one of the $n!$ orderings of their values is equally likely, so $\Pr(X_k$ is the largest$) = 1/n$ for each k, and more generally any event defined by the relative order of the X_i has probability equal to its share of the $n!$ orderings.*

Proof. Ties have probability zero. Permuting the coordinates is a bijection of the sample space that preserves the product measure, and it maps the event "the order is σ" to "the order is $\tau\sigma$"; so all orderings have the same probability, which must be $1/n!$. □

Lemma 1.6 (Sign-reversing involution)**.** *Let S be a finite set, $f\colon S \to \mathbb{R}$, and $\sigma\colon S \to S$ an involution with $f(\sigma(s)) = -f(s)$ whenever $\sigma(s) \neq s$. Then $\sum_{s \in S} f(s) = \sum_{s:\ \sigma(s)=s} f(s)$.*

Proof. The non-fixed points split into two-element orbits $\{s, \sigma(s)\}$, each contributing $f(s) + f(\sigma(s)) = 0$. □

First moves. Pair the terms: k with $n-k$, a subset with its complement, a permutation with its inverse or with its composite with a fixed transposition, a point with its reflection. Ask whether the answer depends on the parameter at all; if the problem gives a specific large number, compute the case with 2 or 3 in its place and see whether the number was ever used. Write the quantity as a sum of indicators and compute one probability. Fix the thing that was random and randomise the thing that was fixed; the probability does not change, and often the second version is the easy one.

Example 1.7. A permutation σ of $\{1, \dots, n\}$, $n \geq 2$, is chosen uniformly at random. Call a position i with $1 \leq i \leq n-1$ a step if $|\sigma(i+1) - \sigma(i)| = 1$. Find the expected number of steps.

Solution. Let N be the number of steps and A_i the event that position i is a step. By linearity $\mathbb{E}N = \sum_{i=1}^{n-1} \Pr(A_i)$. For a fixed i the ordered pair $(\sigma(i), \sigma(i+1))$ is uniform on the $n(n-1)$ ordered pairs of distinct values, because every permutation of the values maps the set of permutations bijectively to itself (transport of uniformity). Exactly $2(n-1)$ ordered pairs of distinct values differ by 1: the pairs $(k, k+1)$ and $(k+1, k)$ for $1 \leq k \leq n-1$. So $\Pr(A_i) = 2(n-1)/(n(n-1)) = 2/n$ for every i, and

$$\mathbb{E}N = (n-1) \cdot \frac{2}{n} = \frac{2(n-1)}{n}.$$

Check at $n = 2$: both permutations have one step, and the formula gives 1. At $n = 3$: the six permutations have step counts $2, 1, 1, 1, 1, 2$, average $4/3$, as the formula says. The events A_i are far from independent, and nothing above needed them to be. □

Example 1.8. For $n \geq 1$ let $\mathrm{fix}(\sigma)$ be the number of fixed points of a permutation σ of $\{1, \ldots, n\}$ and $\mathrm{sgn}(\sigma)$ its sign. Compute

$$T_n = \sum_{\sigma \in S_n} \mathrm{sgn}(\sigma)\, \mathrm{fix}(\sigma).$$

Solution. Write $\mathrm{fix}(\sigma) = \sum_{i=1}^{n} \mathbf{1}[\sigma(i) = i]$ and swap the sums:

$$T_n = \sum_{i=1}^{n} \sum_{\sigma:\ \sigma(i)=i} \mathrm{sgn}(\sigma).$$

Permutations fixing i are permutations of the other $n-1$ elements with the same sign, so each inner sum is $\sum_{\tau \in S_{n-1}} \mathrm{sgn}(\tau)$. For $n - 1 \geq 2$ this is zero: composing with a fixed transposition $(a\, b)$ of two of the $n-1$ elements is a sign-reversing involution with no fixed points. For $n - 1 = 1$ the inner sum is 1, and for $n - 1 = 0$ it is 1 as well (the empty permutation has sign 1). Hence $T_1 = 1$, $T_2 = 2$, and $T_n = 0$ for all $n \geq 3$.

Check at $n = 2$: the identity contributes $+2$, the transposition contributes 0. At $n = 3$: the identity gives $+3$, the three transpositions give -1 each, the two 3-cycles give 0; total 0.

The lesson is in the small cases. The involution that kills the sum needs two elements to swap, and when there are not two the sum survives. A pairing argument always needs its fixed points listed. □

Trap. An independence that looks obvious and is false. Whether the first and second largest of a random sample land in particular positions are not independent events; whether two given positions of a random permutation are fixed points is not independent. The smallest case usually exposes it, so list all outcomes for $n = 3$ before multiplying anything. The opposite trap: a uniformity that is true but was never used. The sum of n fair dice is uniform modulo 6 by the equal-fibres lemma, but modulo 7 it is not (one die gives probability 0 of a multiple of 7), because 7 is not the size of any coordinate's range. A pairing that leaves a fixed point unpaired, as in the second example, gives the wrong answer for exactly the cases where the fixed points exist. Finally, when a quantity does not depend on a parameter, say so and prove it; graders do not accept "by symmetry" without the bijection that realises it.

Ladder.

1. PS 1 #1. Independent draws; the chance they come out in a given order.
2. PS 1 #2. Ways of writing an integer as an ordered sum; a count by a bijection.
3. PS 1 #3. Zero-one matrices with a parity condition on lines; how many.
4. 2018 B1. A set of lattice vectors; remove one so the rest split into two equal-sum halves.
5. 2020 B1. A signed sum over integers by the parity of their binary digits; a residue.
6. PS 1 #12. Passengers taking seats with one rule broken; a probability about the last one.
7. PS 1 #20. Particles on a segment that swap on collision; a time to clear.
8. 2019 A4. A function with vanishing averages over all spheres of one size; must it be zero.

9. 2021 A4. A double integral over a growing disc; the limit.
10. 2023 B3. A random sequence and a longest subsequence of alternating type; its expectation.
11. 2017 A6. Colourings of the edges of a polyhedron with a local condition; a count.

Where it hides. Inside number theory, where a residue count is the same in every class because translation moves the sum by a unit (2020 B1, 2017 B6). Inside probability, where "expected" hides a sum of indicators (2023 B3, 2016 B4). Inside analysis, where an integral over a symmetric region vanishes by an odd reflection (2019 A4, 2021 A4). Inside counting, where a set is split into two equal halves by an involution (2018 B1). Inside recurrences, where a sequence returns to a value because the map that defines it is a bijection of a finite set (2018 B4).